

Mergestack

Exercise 4.5 (pg 225)

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Q10) I proved $\rightarrow$ side of conditional statement $S \Leftarrow R$.

$$\text{Supp } (x, y) \in A \times A \wedge (x, y) \in S \quad \longrightarrow \quad (x, y) \in R$$

$$(x, y) \in S \text{ iff } \exists x \in F (x, y) \in x \times x$$

$$\text{So let } x \in F \wedge (x, y) \in x \times x$$

$$\text{as } F = A/R = \{[a]_R \mid a \in A\}$$

$$\text{so } x \in F \text{ iff } \exists a \in A (x = [a]_R)$$

$$\text{so } ((\exists a \in A \wedge x = [a]_R) \wedge (x, y) \in x \times x) \text{ iff } (x, y) \in [a]_R \times [a]_R$$

$$x \in [a]_R \wedge y \in [a]_R$$

$$a R x \wedge a R y$$

By reflexivity of $R \rightarrow x R x$

By transitivity of $R \rightarrow x R y$

$$\text{so } (x, y) \in R.$$

But there is a simpler cleaner proof.

As it's given that S is determined by F . so by theorem

4.5.6, $A/S = F : A/R$. The proof that $A/R = S$ follows from exercise 9.

Q9) R & S are equivalence relations on A

$$A/R = A/S$$

$$R = S$$

$$\rightarrow \text{Supp } (x, y) \in A \times A.$$

$$\forall (x, y) \in A \times A [x R y \iff x S y]$$

$$\text{Supp } (x, y) \in R.$$

since $x \in A, y \in A$ & $x R y$ By def

of $[x]_R, y \in [x]_R$. By lemma 4.5.5,

We also know that $[x]_R = [y]_R$. so $x \in [y]_R$

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clearly belongs to some class $[x]_R$.

If we can prove that $[x]_R = [x]_S$ we would know that

(*) $x \in y \in [x]_S$

By def $y \in [x]_S$ would mean $y S x$ & by reflexivity of $S \rightarrow x S y$.

so we will give proof that $\forall x \in A ([x]_R = [x]_S)$
let $x \in A$ be arbitrary $\rightarrow [x]_R = [x]_S$
 $x \in [x]_R \Leftrightarrow x \in [x]_S$

This is straightforward for both directions

As both R & S are equivalence relations, so by lemma 4.5.5,

$x \in [x]_R \wedge x \in [x]_S$. (short proof $\rightarrow$ can be omitted)

Now original proof can be done by proving both sides separately or using string of equivalences.

let $(x, y) \in A \times A$ be arbitrary

then $(x, y) \in R$ iff $x \in [y]_R$ iff $x \in [y]_S$ iff $(x, y) \in S$.

Remember it as follows

$(x, y) \in R$ iff x & y belongs to same partition or $[x] = [y]$
we can also say $x \in [y]$ or $y \in [x]$

Q4) (a) Not an equivalence relation because it's not symmetric.
e.g. $2 - 1 = 1 \in \mathbb{N}$ but $1 - 2 = -1 \notin \mathbb{N}$.

(b) It's an equivalence relation. The classes will be

$[x]_R = \{y \mid x - y \in \mathbb{Q}\}$

If we let $x - y = q$ then $y = x - q$.

Since $\mathbb{Q}$ is closed under negation. (A set is closed under negation if $\forall x \in A (-x \in A)$) so we can just write

$$[x]_s = \{x + q \mid q \in \mathbb{Q}\}.$$

All of the rational numbers would fall into one class.

$$[0] = 0 + q \mid q \in \mathbb{Q} = \text{set of Rational numbers} = \mathbb{Q}.$$

Irrational numbers would form countably infinite classes.

e.g. $[\sqrt{2}] = \sqrt{2} + q \mid q \in \mathbb{Q}$ is a class.

No $\sqrt{2} + q \in [x]$. To see why
supp there is a common elem z .

as $z \in [x]$ so $z \in \mathbb{Q}$.

$$z \in [\sqrt{2}] \text{ so } z = \sqrt{2} + q \text{ where } q \in \mathbb{Q}.$$

setting these equal $z = \sqrt{2} + q$

$$\text{Rational} = \sqrt{2} + \text{Rational}$$

$$\sqrt{2} = \text{Rational} - \text{Rational}$$

But the diff of 2 rational numbers is always rational so
this is a contradiction. Hence there are no 2 common elements between
 $[\sqrt{2}]$ and $[\text{Rational}]$

every real number
belongs to exactly
one equivalence class

$$\sqrt{2} - \sqrt{2} = 0 \in \mathbb{Q}$$

$$\text{so } (\sqrt{2}, \sqrt{2}) \in S$$

$$\& \sqrt{2} \in [\sqrt{2}] = \{\sqrt{2} + q \mid q \in \mathbb{Q}\}$$

$$(B) \quad T = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid \exists n \in \mathbb{Z} (y = x \cdot 10^n)\}$$

~~Reflexivity~~ ^{Symmetry} $\rightarrow$ if $x = y \times 10^n$ then $y = \frac{x}{10^n} \rightarrow y = x \times 10^{-n}$
 $0 - n \in \mathbb{Z}$.

Reflexivity $\rightarrow x = x \times 10^0$

Transitivity $\rightarrow x = y \times 10^n$

$$\& y = z \times 10^m \text{ then } y = \frac{x}{10^n}$$

$$\text{so } \frac{x}{10^n} = z \times 10^m$$

$$x = z \times 10^{2n} \text{ where } 2n \in \mathbb{Z}.$$

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So $\sim$ is an equivalence relation.

The classes are like e.g. $[2] = \{0.00 \dots 02, 0.0 \cdot 2, 0.02, \dots, 2, 20, 200, 200 \dots 0\}$

$[x] =$ numbers who have same digits (significant) as x but ~~perhaps~~ decimal point at different places.

Q5) (a) Clearly R is an equivalence relation. The classes are families of parallel lines.

(b) S is not reflexive & transitive so it's not an equivalence relation.

(c) The T is of those lines that intersect x & y axis on same points. Clearly it's an equivalence relation. The set of all lines that pass through origin form an equivalence class. All other ~~other~~ classes just contain a single line (like the relation

(a))

Q7) $S = \{ (s, t) \in T \times T \mid \text{Triangles } s \text{ \& } t \text{ are similar} \}$

Reflexivity: Every triangle is similar to itself.

Symmetry & transitivity is also easy.

e.g. for transitivity, if all angles of triangle A are equal to angles of ΔB & all $\angle$ s of ΔB are = $\angle$ s of ΔC then all $\angle$ s of ΔA are = ΔC .

if $\{a_1, a_2, a_3\}$ be set of angles of ΔA & $\{b_1, b_2, b_3\}$ be set of angles of ΔB & $\{c_1, c_2, c_3\}$ be set of angles of ΔC then $A \sim B$ & $B \sim C$ then $A \sim C$ as for any three sets, if $A \sim B$ & $B \sim C$ then $A \sim C$.

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Q9) correct proof of $\forall x ([x]_R \subseteq [x]_S)$

Let x be arbitrary s.t. $x \in A$.

$$[x]_R = [x]_S$$

$$\forall a \in A [a \in [x]_R \leftrightarrow a \in [x]_S]$$

Let $a \in A$ be arbitrary. s.t.

$$\rightarrow a \in [x]_R$$

$$\therefore x R a$$

$$\rightarrow a \in [x]_S$$

iff $a S x$

R & S are equivalence relations on A

$$A/R = A/S$$

$$\{[x]_R \mid x \in A\} = \{[x]_S \mid x \in A\} \text{ so } \bigcup_{x \in A} [x]_R = \bigcup_{x \in A} [x]_S$$

as $a \in [x]_R$ means $\exists x \in A (a \in [x]_R)$

The only way to proceed is supposing $x \in A$
showing $A/R = A/S$

as $[x]_R \in A/R$ so $[x]_R \in A/S$

which means $\exists y \in A ([x]_R = [y]_S)$

$[x]_R = [y]_S$ means $x = y$ ($\because$ as $x \in [x]_R$ & $x \in [y]_S$

so replacing $[y]_S$ by $[x]_R$ & similar reasoning)

$[x]_S$ as $x = y$, we get $[x]_R = [x]_S$

Q11) (b) $\mathbb{Z}/\equiv_2 = \{ \{ \dots, -4, -2, 0, 2, 4, \dots \}, \{ \dots, -3, -1, 1, 3, \dots \} \}$
= { set of even integers, set of odd integers }

so there are 2 equivalence classes.

Clearly the union of both classes = $\mathbb{Z}$

$\mathbb{Z}/\equiv_3$ has 3 equivalence classes $\{ 3n \mid n \in \mathbb{Z} \} = \{ \dots, -3, 0, 3, \dots \}$

$$\{ \dots, -7, -4, -1, 2, 5, 8, \dots \} = \{ 3n-1 \mid n \in \mathbb{Z} \}$$

$$\{ \dots, -5, -2, 1, 4, 7, 10, \dots \} = \{ 3n+1 \mid n \in \mathbb{Z} \}$$

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$$Q12) \forall n \in \mathbb{Z} [n^2 \equiv_4 0 \vee n^2 \equiv_4 1]$$

supp $n \in \mathbb{Z}$

$$4 \mid n^2 - 0 \vee 4 \mid n^2 - 1$$

$$\exists k \in \mathbb{Z} (n^2 = 4k) \vee \exists k \in \mathbb{Z} (n^2 - 1 = 4k)$$

Case 1) supp n is even. Then $n = 2l$ for $l \in \mathbb{Z}$

$$n^2 = (2l)^2 = 4l^2. \text{ Clearly } l^2 \in \mathbb{Z} \text{ so } k = l^2 \text{ works.}$$

$$(\because n^2 = 4l^2 = 4(l^2))$$

Case 2) n is odd. Then $n = 2l+1$ for $l \in \mathbb{Z}$.

$$n^2 = 4l^2 + 4l + 1 = 4(l^2 + l) + 1 = 4(k) + 1$$

$$n^2 - 1 = 4(l^2 + l)$$

$$\text{so taking } k = l^2 + l \text{ works } \& \ l^2 + l \in \mathbb{Z}$$

$$Q13) m \in \mathbb{Z}$$

~~we have to prove~~

we have to prove that

$$\text{supp } a, a^3, b, b^3 \in \mathbb{Z}$$

$$(a^3 + b^3 \equiv_m a + b) \wedge$$

$$\text{supp } a^3 \equiv_m a$$

$$(a^3 b^3 \equiv_m ab)$$

$$\& \ b^3 \equiv_m b$$

$$\therefore \exists k \in \mathbb{Z} (a^3 + b^3 - a - b = km)$$

$$\text{First } a^3 \equiv_m a$$

$$\wedge \exists k \in \mathbb{Z} (a^3 b^3 - ab = km)$$

$$\text{means } a^3 - a = im \quad (3)$$

$$\text{or } a(a^2 - 1) = im \quad (1)$$

$$\text{Similarly } b^3 \equiv_m b \text{ means } b(b^2 - 1) = im \quad (2)$$

$$\text{add } (1) \& (2), \ a^3 + b^3 - a - b = im + im$$

$$= (l+i)m \text{ so } k = l+i \text{ works}$$

now second part

The key is not multiplying but plugging values of

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$a^3 \notin b^3$. The first part can also be solved like this.

from (3), we get $a^3 = a + lm$

& similarly $b^3 = b + im$.

$$\text{Now } a^3 \odot b^3 - ab = (a + lm) \odot (b + im) + ab$$

$$= ab + aim + blm + lim^2 \odot ab$$

$$= aim + blm + lim^2 = m(ai + bl + lim)$$

so $k = ai + bl + lim$ works & hence

$$m \mid a^3 b^3 - ab. \text{ so } a^3 b^3 \equiv_m ab.$$

($\because$ Note $m \in \mathbb{Z}^+$ so $lim \in \mathbb{Z}$)

(Q14) R is equivalence relation on A

$$B \subseteq A$$

$$S = R \cap (B \times B)$$

(a)

S is equivalence on B

Reflexivity

$$\text{supp } b \in B$$



$$(b, b) \in S$$

as $B \subseteq A$ so $b \in A$. As $R \subseteq A \times A$ & R is equivalence so

$(b, b) \in R$ and since $(b, b) \in B \times B$ so $(b, b) \in R \cap (B \times B)$

$$\text{so } (b, b) \in S$$

Symmetry

$$\text{supp } (b, c) \in S$$



$$(c, b) \in S$$

$$\text{so } (b, c) \in B \times B$$

~~$b \in B$ & $c \in B$ & as $b \in A$ so $b \in A$ & $c \in A$ so $(b, c) \in A \times A$~~

As $S = R \cap (B \times B)$ so $(b, c) \in R$ and as R is equivalence

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so $(c, b) \in R$ & clearly $(c, b) \in B \times B$ so $(c, b) \in S$

Transitivity

Supp $(b, c) \in S \wedge (c, d) \in S \longrightarrow (b, d) \in S$

so $(b, c) \in R$ & $(c, d) \in R$

since R is equivalence relation so by transitivity of R ,
 $(b, d) \in R$ & clearly $(b, d) \in B \times B$

so $(b, d) \in S$.

(b) $\forall x \in B ([x]_S = [x]_R \cap B)$

Supp $x \in B$ — ①

$[x]_S = [x]_R \cap B$

$\longrightarrow$ Supp $a \in [x]_S \longrightarrow a \in [x]_R \cap B$

iff $(a, x) \in S$ or $a \sim x$

so $(a, x) \in R \cap (B \times B)$

so $(a, x) \in R \wedge (a, x) \in B \times B$

so $a \in [x]_R$ and $a \in B$

so $a \in [x]_R \cap B$

Supp $a \in [x]_R \cap B$

$\longrightarrow a \in [x]_S$

so $(a, x) \in R \wedge a \in B$

$= (a, x) \in S$

$(a, x) \in R \wedge a \in B$

if we can prove that $x \in B$ we can conclude $(a, x) \in B \times B$

& as $S = R \cap (B \times B)$ so $(a, x) \in S$ & so $a \in [x]_S$

so our goal is $x \in B$

By ① $\rightarrow$ its clearly given. we have proved

Q15) $B \subseteq A$

$$R = \{ (X, Y) \in P(A) \times P(A) \mid X^2 Y \subseteq B \}$$

$$\text{where } X^2 Y = (X \setminus Y) \cup (Y \setminus X)$$

$$\text{or } X^2 Y = (X \cup Y) \setminus (X \cap Y)$$

(a)

R is equivalence relation on $P(A)$

Reflexivity:

Supp $X \in P(A)$



$$(X, X) \in R$$

$$X^2 X \subseteq B$$

$$(X \cup X) \setminus (X \cap X) \subseteq B$$

$$X \setminus X \subseteq B$$

$$\emptyset \subseteq B$$

Clearly $\emptyset \subseteq$ every set.

Symmetry:

Supp $(X, Y) \in R$



$$(Y, X) \in R$$

$$\text{iff } X^2 Y \subseteq B$$

$$Y^2 X \subseteq B$$

$$\text{iff } (X \cup Y) \setminus (X \cap Y) \subseteq B$$

$$\text{iff } (Y \cup X) \setminus (Y \cap X) \subseteq B \quad (\because \text{commutative law})$$

$$\text{iff } Y^2 X \subseteq B$$

Transitivity

Supp $(X, Y) \in R \wedge (Y, Z) \in R$

$$\longrightarrow (X, Z) \in R$$

$$X^2 Y \subseteq B \text{ (1)} \wedge Y^2 Z \subseteq B \text{ (2)}$$

$$X^2 Z \subseteq B$$

$$\forall x [x \in X^2 Y \rightarrow x \in B] \quad (1)$$

$$(X \cup Z) \setminus (X \cap Z) \subseteq B$$

$$\forall x [x \in Y^2 Z \rightarrow x \in B] \quad (2)$$

$$\forall x [x \in (X \cup Z) \setminus (X \cap Z) \rightarrow x \in B]$$

$$\forall x [(x \in X \vee x \in Y) \wedge (x \notin X \cup x \notin Y) \rightarrow x \in B] \quad (1)$$

$$\forall x [(x \notin Y \vee x \in Z) \wedge (x \notin Y \vee x \notin Z) \rightarrow x \in B] \quad (2)$$

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Supp $x \in (X \cup Z) \setminus (X \cap Z) \longrightarrow x \in B$

iff $(x \in X \vee x \in Z) \wedge (x \notin X \vee x \notin Z) \rightarrow (3)$

Case 1) $x \in X$.

Then as $x \notin X \vee x \notin Z$ we can conclude $x \notin Z$.

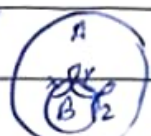
~~So $x \in X$ so $x \notin X \cup Z$~~

and as $x \notin Z$ so $x \notin X \cap Z$

So $x \in (X \cup Z) \setminus (X \cap Z)$

So

For better understanding of transitivity. Make a venn diagram



Case 1.1) Supp $x \in Y$

Then from (2), we can conclude $x \in B$.

Case 1.2) Supp $x \notin Y$, then from (1) we can conclude $x \in B$.

Case 2) $x \in Z$.

Then from (2), we can conclude $x \notin X$.

Case 2.1) Supp $x \in Y$.

Then from (1), we can conclude $x \in B$.

Case 2.2) Supp $x \notin Y$

Then from (2), we can conclude $x \in B$.

Since each of these cases are exhaustive, we conclude $x \in B$.

Since x was arbitrary elem of $X \cap Z$ so $X \cap Z \subseteq B$ thus $(X \cap Z) \subseteq B$.

For this statement also make a venn diagram

(b) $\forall x \in P(A) [\exists! y \in [X]_R (y \cap B) = \emptyset]$

Note that an equivalence class $[x]_R$ is a family of sets

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$$\forall x \in P(A) [\exists y \in [x]_R (Y \cap B) = \emptyset \wedge \forall y' \forall y_2 \in [x]_R ((Y_1 \cap B) = \emptyset \wedge (Y_2 \cap B) = \emptyset) \longrightarrow Y_1 = Y_2]$$

First Existence

$$\text{Supp } x \in P(A) \longrightarrow \exists y \in [x]_R (Y \cap B) = \emptyset$$

$$\text{Let } Y = x \cap B$$

then first to show that $y \in [x]_R$ we need to show that $(Y, x) \in R$ or $y \cap x \in B$

$$\text{Supp } x \in Y^2 X \longrightarrow x \in B$$

$$\text{or } x \in ((x \cap B) \cup x) \setminus ((x \cap B) \cap x)$$

$$\text{This expands to } x \in x \cap B \vee x \in x$$

$$\text{all } x \notin (x \cap B) \vee x \notin x$$

which further expands to

$$(x \in x \wedge x \notin B) \vee x \in x \quad \text{--- (1)}$$

$$\text{and } x \notin x \vee x \in B \quad \text{--- (2)}$$

Case 1) $x \notin x$. The free (1), we can conclude

$$x \in x \wedge x \notin B \text{ which is contradiction as } x \notin x. \text{ so}$$

this case is not possible.

Case 2) $x \in B$. The goal is proved. We basically ruled out case that $x \notin x$.

$$\text{So now we know } Y = x \cap B \in [x]_R. \text{ We need to show that } (Y \cap B) = \emptyset$$

$$\exists x (x \in Y \wedge x \in B)$$

Proof by contradiction. Supp $x \in Y \wedge x \in B$

$$\text{not } Y \text{ means } x \in x \cap B \text{ means } x \in x \wedge x \notin B \text{ which is contradiction}$$

$$\text{over } x \in B. \text{ So } Y \cap B = \emptyset$$

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Uniqueness

To prove uniqueness, it's better to use other method

$$\exists x [P(x) \wedge \forall y (P(y) \rightarrow y = x)]$$

In our problem $\exists Y_1 \in [X]_R [Y_1 \cap B = \emptyset \wedge \forall Y_2 \in [X]_R (Y_1 \cap B = \emptyset \Rightarrow Y_1 = Y_2)]$

We already showed that $Y = X \setminus B$ works.

Now to see uniqueness, supp

$$Y_1 \in [X]_R \text{ \& } Y_1 \cap B = \emptyset \quad \rightarrow \quad Y_1 = Y_2 \text{ or } Y_1 = X \setminus B$$

if $(Y_1, X) \in R$

$$\text{if } \forall x [(x \in Y_1 \vee x \in X) \wedge (x \notin Y_1 \vee x \notin X) \rightarrow x \in B] \quad \textcircled{1}$$

$$\text{and } \forall x [x \in Y_1 \rightarrow x \notin B] \quad \textcircled{2} \quad \because Y_1 \cap B = \emptyset$$

$\rightarrow$ Supp $y \in Y_1$

so by $\textcircled{2}$, $y \notin B$

$$y \in Y_1$$

$$y \in X \setminus B$$

$$y \in X \wedge y \notin B$$

Proof by contradiction

Supp $y \notin X \setminus B$. Then $y \notin X$ or $y \notin B$

We know $y \notin B$ so $y \notin X$.

But then as $y \in Y_1$, by $\textcircled{1}$, we can conclude $y \in B$ which

is contradiction over $y \notin B$. So $y \in X$. Hence $y \in X \setminus B = Y$

$\leftarrow$ Supp $y \in X \setminus B$

$$\text{then } y \in X \wedge y \notin B$$

$$y \in Y_1$$

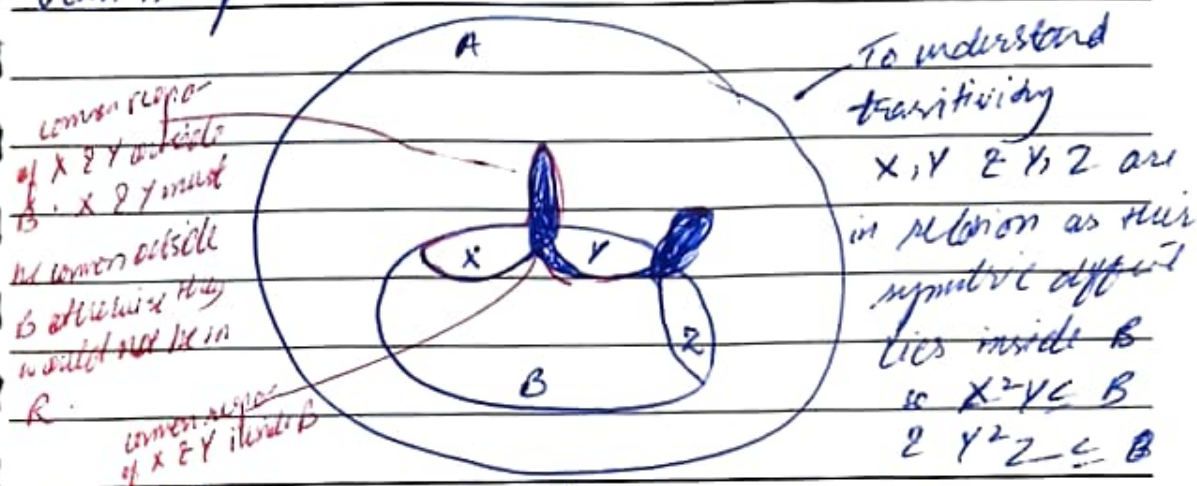
Proof by contradiction

Supp $y \notin Y_1$. As $y \in X$ so from $\textcircled{1}$, we can conclude $x \in B$

which is contradiction over $y \notin B$. So $y \in Y_1$.

Hence $Y_1 = Y_2$

Venn Diagram To understand Q: 15 (a) Transitivity



To see $X \cap Z$, we considered cases as an arbitrary x can belong to either X or Z (see the proof). We consider whether $x \in Y$ or $x \notin Y$. In both cases if we see Venn diagram e.g. for $x \in X$, we know $x \notin Z$. So x is in $Y \cap Z$ so $x \in B$. As $x \in B$ & $x \in X$ & $x \in Y$ so x must lie in common region of X & Y that is inside B . Similar logic is for other cases as well.

Venn Diagram to understand Q: 15 (b)

Consider the same Venn diagram. For any $x \in P(A)$ or any $x \in A$ we want to find a subset Y such that $Y \cap B = \emptyset$. First so that we only have X, Y, B & $\{X\}_R$ here on which Y can depend or can be made from. To make $Y \cap B = \emptyset$, we must not have any B inside Y . So something like $\setminus B$ will work. Consider X in Venn diagram. Clearly we should have $Y \in \{X\}$ so it might be good to try $X \setminus B$. If we remove B from X it will not have anything common with B so $(X \setminus B) \cap B = \emptyset$. So Venn diagrams are a great way to find these kind of sets.

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In diagram, we can also see that $x, y \in Z$ are in R . So they must belong to same equivalence class. Taking out B from x in venn diagram leaves part of x that is common with y & so this must also belong to $[x]_R$. If we focus on logic symbols in proof & venn diagram, everything will make sense.

Q16) F is partition of A
 G is partition of B
 $A \cap B = \emptyset$

$F \cup G$ is partition of $A \cup B$

$$1) \cup(F \cup G) = A \cup B$$

→ This means $\exists x \in (F \cup G) (\forall x \in x) \longrightarrow x \in A \cup B$

Case 1) $x \in F$

Then as $x \in x$ so $x \in \cup F$ & since F is partition of A
so $\cup F = A$ so $x \in A$.

Case 2) is similar for G .

2) $F \cup G$ is pairwise disjoint

$$\text{Supp } x \in F \cup G \wedge y \in F \cup G$$

$$\text{Supp } x \neq y$$

$$\longrightarrow x \cap y = \emptyset$$

$$\text{Supp } x \in x \wedge x \in y \text{ or } x \in x \cap y \longrightarrow \text{Contradiction}$$

There are 4 cases 1) $x \in F$ & $y \in F$

then since F is pairwise disjoint $x \cap y = \emptyset$.

Case 2) $x \in F \wedge y \in G$

Supp $x \cap y \neq \emptyset$ so $x \in x \wedge x \in y$ then $x \in \cup F = A$ so $x \in A$ &
 $y \in \cup G = B$ so $y \in B$. This is contradiction over $A \cap B = \emptyset$.

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Case 3) $x \in G \wedge y \in F$.

similar to case 2.

Case 4) $x \in G \wedge y \in G$.

similar to case 1

$$3) \forall x \in F \cup G (x \neq \emptyset)$$

supp $x \in F \vee x \in G$.

$x \neq \emptyset$

Case 1) $x \in F$.

since F is partition of A so F is non empty. Thus $x \neq \emptyset$

Case 2) is similar

Q17) (a) easy. Transitivity lead to 4 cases out of which 2 aren't possible.

(b) $\forall x \in A, [x]_{R \cup S} = [x]_R$

supp $x \in A$

$$[x]_{R \cup S} = [x]_R$$

→ supp $y \in [x]_{R \cup S}$. Then $(y, x) \in R \cup S$.

If $(y, x) \in S$ then $x \in B$ as S is a relation on B but we know that $x \in A$ so its contradiction over fact that $A \cap B$ is empty disjoint so $(y, x) \notin S$. so $(y, x) \in R$ so $y \in [x]_R$.

← easy case.

The second part is similar

(c) $(A \cup B) / (R \cup S) = (A/R) \cup (B/S)$

→ supp $x \in (A \cup B) / (R \cup S)$ → $x \in A/R \vee x \in B/S$

$$x \in \{ [x]_{R \cup S} \mid x \in A \cup B \}$$

$$\exists x \in A (x = [x]_R) \vee$$

$$\exists x \in B (x = [x]_S)$$

$$\exists x \in B (x = [x]_S)$$

Case 1) $x \in A \wedge x = [x]_R$

Case 2) If $x = [x]_S$ then as $x \in [x]_S$ so $x \in S$

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so $x_0 \in B$ as $S \subseteq B \times B$. This is contradiction over $A \cap B = \emptyset$.

so $X \neq \{x_0\}_S$ so $X = \{x_0\}_R$. & goal follows.

Case 2 is similar

← Side is easy.

Q18) F & G are partitions of A

$$F, G = \{Z \in P(A) \mid Z \neq \emptyset \wedge \exists X \in F \exists Y \in G (Z = X \cap Y)\}.$$

Goal: $F \cdot G$ is partition of A .

1) $\bigcup (F \cdot G) = A$.

→ Supp $x \in \bigcup (F \cdot G) \longrightarrow x \in A$

$\exists X \in F \cdot G (x \in X)$ means

$$X \in P(A) \wedge X \neq \emptyset \wedge \exists Y \in F \exists Z \in G (X = Y \cap Z)$$

& $x \in X$. Since $X \in P(A)$ so $X \subseteq A$. As $x \in X$ so $x \in A$.

2) $F \cdot G$ is pairwise disjoint.

$$\forall X \in F \cdot G \forall Y \in F \cdot G [X \neq Y \longrightarrow X \cap Y = \emptyset]$$

Supp $X \in F \cdot G \wedge Y \in F \cdot G \wedge X \neq Y \longrightarrow X \cap Y = \emptyset$

By contradiction, assume $X \cap Y \neq \emptyset$ so $x \in X \cap Y$

we can see that $x \in A$.

$X \in F \cdot G$ means $X \subseteq A \wedge X \neq \emptyset \wedge \exists X_1 \in F \exists X_2 \in G (X = X_1 \cap X_2)$

$Y \in F \cdot G$ means $Y \subseteq A \wedge Y \neq \emptyset \wedge \exists Y_1 \in F \exists Y_2 \in G (Y = Y_1 \cap Y_2)$

As $x \in X$ & $X = X_1 \cap X_2$ so $x \in X_1 \wedge x \in X_2$

Similarly $x \in Y_1 \wedge x \in Y_2$. But $Y_1 \in F$ & $X_1 \in F$ so this

is contradiction over fact that F is pairwise disjoint.

so $X \cap Y = \emptyset$

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$$3) \forall x \in F \cap (x \neq \emptyset)$$

This follows by definition as only those sets are inside $F \cap$ which are non empty

Q19) $F = \{R, R^+, \{0\}\}$ $G = \{Z, R \setminus Z\}$. Clearly these are partitions of R .

$$F \cdot G = \{Z^-, R^- \setminus Z, Z^+, R^+ \setminus Z, \{0\}\}$$

Note that $F \cdot G$ is more refined & filtered.

Q20) $R \cap S$ are equivalence relations on A

$$T = R \cap S$$

(a)

T is equivalence on A .

Reflexivity

Supp $x \in A$. As $R \cap S$ are equivalence so xRx & xSx

$$\Rightarrow (x, x) \in R \cap S = T$$

Symmetry

$$\text{Supp } (x, y) \in T = R \cap S$$

$\Rightarrow x, y \in R \cap S$ by their symmetry $(y, x) \in R \cap S \Rightarrow yTx$

Transitivity

$$(x, y) \in T \text{ \& } (y, z) \in T$$

easy again.

(b)

$$\forall x \in A ([x]_T = [x]_R \cap [x]_S)$$

$$\text{Supp } x \in A \text{ then } [x]_T = [x]_{R \cap S} = [x]_R \cap [x]_S$$

Replacing T by $R \cap S$ & any $y \in [x]_{R \cap S}$ means $y \in [x]_R \cap [x]_S$

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$$(L) \quad AIT = (A/R) \circ (A/S)$$

$$\rightarrow \text{supp } X \in AIT \rightarrow X \in (A/R) \circ (A/S)$$

$$\text{so there exists } X \in P(A) \wedge X \neq \emptyset \wedge \exists Y \in (A/R) \exists Z \in (A/S)$$

$$\text{some } x \in A \text{ such that } (x = Y \cap Z)$$

$$X = [x]_T \quad \text{--- ①}$$

$$\text{iff } X = [x]_R \cap [x]_S \quad \text{by part (b) --- ②}$$

$$\text{by def of } AIT, X \subseteq A \text{ so } X \in P(A).$$

$$\text{as } X = [x]_T \text{ \& } x \in [x]_T = X \text{ so } X \neq \emptyset.$$

$$\text{by ① } X = \{y \mid (y, x) \in T = R \cap S\}$$

$$\text{by translating our goal, we can see the proof. As } \cancel{x \in A} \text{ \& } x$$

$$[x]_R \in A/R \text{ so let } Y = [x]_R. \text{ Similarly let } Z = [x]_S$$

$$\text{as } [x]_S \in A/S. \text{ \& } X = [x]_R \cap [x]_S = Y \cap Z.$$

$$\text{For } X \neq \emptyset \text{ goal, we could just have used reasoning that}$$

$$\text{as } AIT \text{ is partition \& } X \in AIT \text{ so } X \neq \emptyset.$$

$$\leftarrow \text{supp } X \in (A/R) \circ (A/S) \rightarrow X \in AIT$$

$$\text{Then } \exists Y \in (A/R) \exists Z \in (A/S) (X = Y \cap Z) \Leftrightarrow \exists x \in A (X = [x]_R \cap [x]_S)$$

$$\text{let } Y_0 \in A/R \text{ \& } Z_0 \in A/S \text{ \& } X = Y_0 \cap Z_0 \Leftrightarrow \exists x \in A (X = [x]_R \cap [x]_S)$$

$$\text{iff } \exists y \in A (Y_0 = [y]_R) \text{ \& } \exists z \in A (Z_0 = [z]_S)$$

$$\text{so } X = Y_0 \cap Z_0 = [y]_R \cap [z]_S$$

$$\text{as } y \text{ \& } z \in A \text{ \& } X = Y \cap Z = [y]_R \cap [z]_S. \text{ If we can prove}$$

$$y = z, \text{ we did it done. This is not possible}$$

$$\text{As } X \in (A/R) \circ (A/S) \text{ so } X \neq \emptyset.$$

$$\text{so } \exists x \in X$$

$$\text{As } X = [y]_R \cap [z]_S \text{ so } x \in [y]_R \wedge x \in [z]_S \text{ so}$$

$$[x]_R = [y]_R \text{ \& } [x]_S = [z]_S. \text{ so } X = [x]_R \cap [x]_S$$

$$\text{(clearly } x \in A \text{ as } X \subseteq A \text{ \& } x \in X.$$

Q21) F is partition of A & G is partition of B .

$U \in A$ & F is pairwise disjoint & $\forall x \in F (x \neq \emptyset)$. Similar for G .

$$F \otimes G = \{ Z \in P(A \times B) \mid \exists x \in F \exists y \in G (Z = x \times y) \}$$

PROVE $F \otimes G$ is partition of $A \times B$.

1) $\bigcup F \otimes G = A \times B$

$\rightarrow \text{Supp}((a,b) \in F \otimes G) \rightarrow (a,b) \in A \times B.$

~~$\text{If } (a,b) \in P(A \times B) \wedge \exists x \in F \exists y \in G ((a,b) = x \times y)$~~

~~$\rightarrow (a,b) \in A \times B$ so $(a,b) \in$~~

$\exists x \in F \otimes G ((a,b) \in x)$

so $x \in P(A \times B) \rightarrow x \subseteq A \times B$ so $(a,b) \in A \times B.$

2) $\text{Supp } x \in F \otimes G \wedge y \in F \otimes G \wedge x \neq y \rightarrow x \cap y = \emptyset$

$x \in P(A \times B) \wedge \exists x_1 \in F \exists x_2 \in G (x = x_1 \times x_2)$

$y \in P(A \times B) \wedge \exists y_1 \in F \exists y_2 \in G (y = y_1 \times y_2)$

$\text{Supp } x \cap y \neq \emptyset$ so $(a,b) \in x \cap y$

Since $(a,b) \in x$ so $(a,b) \in x_1 \times x_2$ & similarly $(a,b) \in y_1 \times y_2$

so $(a,b) \in x_1 \cap y_1$ which is contradiction over fact that

F is pairwise disjoint

3) ~~$\text{Supp } x \in F \otimes G$~~

$x \neq \emptyset$

This means $\exists x \in F \exists z \in G (x = x \times z)$

As F & G are partitions, so $x \neq \emptyset \in F \neq \emptyset$ so $x \times z = x \neq \emptyset$

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$$Q22) F = \{R^-, R^+, \{0\}\}$$

$$F \otimes F = \{R^- \times R^-, R^- \times R^+, R^- \times \{0\}, R^+ \times R^-, R^+ \times R^+, R^+ \times \{0\}, \{0\} \times R^-, \{0\} \times R^+, \{0, 0\}\}$$

In geometric terms, there are 4 quadrants, +ve & -ve x & y axis & origin.

Q23) (a) ^{Reflexivity} ~~symmetry~~ $\rightarrow$ follows from ref of R & S.

Symmetry $\rightarrow$

$$\text{if } ((a, b), (a', b')) \in T$$

$$\text{so } a R a' \text{ \& } b S b' \text{ so } a' R a \text{ \& } b' S b. \text{ Local}$$

clearly follows.

Transitivity:

$$((a, b), (a_1, b_1)) \in T$$

$$\longrightarrow ((a, b), (a_2, b_2)) \in T$$

$$((a_1, b_1), (a_2, b_2)) \in T$$

$$\text{as } a R a_1 \text{ \& } b S b_1$$

$$a_1 R a_2 \text{ \& } b_1 S b_2$$

$a_1 R a_2 \text{ \& } b_1 S b_2$ — Local follows from transitivity of R & S.

$$(b) a \in A, b \in B$$

$$[(a, b)]_T = [a]_R \times [b]_S$$

$$(a, b) \in [(a, b)]_T \text{ iff } ((a, b), (a, b)) \in T$$

$$\text{iff } a R a \text{ \& } b S b \text{ iff } a \in [a]_R \text{ \& } b \in [b]_S$$

$$(c) (A \times B) / T = (A / R) \otimes (B / S)$$

$$\longrightarrow \text{LHS } \times \text{ RHS}$$

$$x \in \text{LHS}$$

$$\text{so } \exists (a, b) \in A \times B \text{ s.t. } (a, b) \in [(a, b)]_T \quad \text{--- (1)}$$

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To prove: $X \in P(A \times B) \wedge \exists Y \in A/R \exists Z \in B/S (X = Y \times Z)$

as $X \in (A \times B) / T$ so $X \subseteq A \times B$ so $X \in P(A \times B)$

By ①

$X = \{(a, b)\}_T = [a]_R \times [b]_S \Rightarrow$ by part (b)

as $[a]_R \in A/R$ & $[b]_S \in B/S$ so goal follows.

$\leftarrow$ Suppose $X \in (A/R) \otimes (B/S)$ $\rightarrow X \in (A \times B) / T$
means $X \in P(A \times B)$ $\exists (a, b) \in A \times B (X = \{(a, b)\}_T)$
 $\exists Y \in (A/R) \exists Z \in (B/S) (X = Y \times Z)$ $\exists (a, b) \in A \times B (X = [a]_R \times [b]_S)$
means $\exists a \in A (Y = [a]_R)$
 $\exists b \in B (Z = [b]_S)$
Clearly $a \in A$ & $b \in B$ so $(a, b) \in A \times B$.
& $X = Y \times Z = [a]_R \times [b]_S$

~~Ex 9.4~~ The pattern b/w $\mathcal{Q}_{1,2,2,0}$ & $\mathcal{Q}_{2,1,2,3}$ is simple.
It was pointed out if they have 2 ~~eq~~ equivalence relations
(or partitions, one can determine other) on some set. Then
they can make a new partition or equivalence relation (one
will determine other). And it was pointed the pattern &
defined the partition & corresponding relation or vice versa.
A key thing is: An equivalence relation makes partition
& vice versa.

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(Q24) R is relation on A .

S is equivalence relation on A

R is compatible with S if

$$\forall x \in A \forall y \in A \forall x_1 \in A \forall y_1 \in A [(x S x_1 \wedge y S y_1) \rightarrow (x R y \leftrightarrow x_1 R y_1)]$$

(a) R is compatible with S

$$\exists T \subseteq (A/S) \times (A/S) [\forall x \in A \forall y \in A$$

$$([x]_S, [y]_S) \in T \iff x R y]$$

By doing 2 examples, we saw that because S is equivalence, so every $(x, x) \in S$. Due to this, we were getting every possible pairs in R as well. Example, since $(1, 1), (2, 2) \in S$ so $(1, 2) \& (2, 1)$ will belong to R along with $(1, 1) \& (2, 2)$ as R is compatible with S . This leads to defining T as follows because for any $[x]_S, [y]_S$ $x \& y$ were in R i.e. $x R y$.

$$\text{so let } T = \{ \overset{x, y}{([x]_S, [y]_S)} \in (A/S) \times (A/S) \mid \exists x \in X \exists y \in Y (x R y) \}$$

$$\rightarrow \text{supp}([x]_S, [y]_S) \in T \rightarrow x R y$$

clearly $x \in [x]_S \& y \in [y]_S$

so $x S x \& y S y$

as R is compatible with S , so $x R y$.

$\leftarrow$ supp $x R y$

(clearly $x \in [x]_S \& y \in [y]_S$

$\&$ we know $x R y$.)

$$[x]_S \neq [y]_S$$

$$\exists x \in [x]_S \exists y \in [y]_S (x R y)$$

if we let so $y_0 \in [y]_S$ here $x R y_0$. By compatibility of R , we can still prove $x R y$

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$$(b) T \subseteq (A/S) \times (A/S)$$

~~Goal~~

Goal

$$\forall x, y \in A ([x]_S, [y]_S) \in T \rightarrow xRy \quad \text{--- } \textcircled{0}$$

R is compatible with S

Using def of compatibility

$$\text{Supp } x_1 \in A, y_1 \in A, x_1 \in A \text{ \& } y_1 \in A$$

$$\text{Supp } x \in x_1 \wedge y \in y_1$$

$$\rightarrow (xRy \rightarrow x_1Ry_1)$$

$$\rightarrow \text{Supp } xRy$$

x, y_1

By $\textcircled{1}$, we can conclude

$$([x]_S, [y]_S) \in T$$

$$\text{Since } x \in x_1 \text{ so } x_1 \in [x]_S$$

$$\text{Similarly } y_1 \in [y]_S$$

$$\text{so } [x_1]_S = [x]_S \wedge [y_1]_S = [y]_S$$

$$\text{so } ([x_1]_S, [y_1]_S) \in T \text{ and as } x_1, y_1 \in A \text{ so by } \textcircled{0},$$

we can conclude x_1Ry_1

$$\leftarrow \text{Supp } x_1Ry_1$$

xRy

similar reasoning

Q25) (a) is ~~straight~~ easy proof

$$(b) \exists T \subseteq (A/S) \times (A/S) \forall x, y \in A ([x]_S, [y]_S) \in T \rightarrow xRy$$

$$\text{Let } T = \{(x, y) \in A/S \times A/S \mid \exists x' \in x \exists y' \in y (x'Ry')\}$$

$$\text{Supp } x, y \in A$$

$$\rightarrow \text{Supp } ([x]_S, [y]_S) \in T$$

$$\rightarrow xRy$$

$$x \in [x]_S \text{ \& } y \in [y]_S \rightarrow x \in x \text{ \& } y \in y$$

To use ~~lemma~~ 24(a), we can just prove that R is compatible with S , then required conclusion follows from 24(a)

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To prove that R is compatible with S .

Supp $x, x_1, y, y_1 \in A$.

Supp $x S x_1 \wedge y S y_1$

$\rightarrow (x R y \leftrightarrow x_1 R y_1)$

$\rightarrow$ Supp $x R y$ - Also $y R x$ by reflexivity of R

Since $S = R \cap R^{-1}$

so $x R x_1$ as $x S x_1$ & similarly $y R y_1$

by reflexivity of R , $x_1 R x$ & by transitivity of R , we get $x_1 R y$. Again by transitivity of R as $y R y_1$,

so $x_1 R y_1$.

$\leftarrow$ A similar argument.

So R is compatible with S &

(c) $T = \{ (x, y) \in A/S \times A/S \mid \exists x \in X \exists y \in Y (x R y) \}$.

This is partial order on A/S

Reflexivity.

Supp $X \in A/S$

$(X, X) \in T$

~~QED~~ $\exists x_0 \in A (X = [x_0]_S)$

$\exists x \in X \exists y \in X (x R y)$

but $x = y = x_0$.

As $[x_0]_S = X$ & $x_0 \in [x_0]_S$ so $x_0 \in X$.

& since $x_0 \in A$ & R is reflexive, so $x_0 R x_0$.

Antisymmetry $\forall x, y \in A/S \quad \forall x \in X \forall y \in Y \quad [(x, y) \in T \wedge (y, x) \in T \rightarrow x = y]$

Supp $X \in A/S$ & $Y \in A/S$

Supp $X \geq Y \in T \wedge Y \geq X \in T$

$X = Y$

~~then $X = Y$~~

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$$\exists x \in X \exists y \in Y (xRy)$$

$$\exists y \in Y, \exists x \in X (yRx)$$

$$\text{so } x_1 \in X \ y_1 \in Y (x_1 R y_1)$$

$$y_2 \in Y \ x_2 \in X (y_2 R x_2)$$

$$\text{Since } R \text{ is } \text{so } \exists x \in A (X = [x]_S) \checkmark$$

$$\text{Similarly } \exists y \in A (Y = [y]_S)$$

$$\text{so } x_0 \in A \wedge X = [x_0]_S$$

$$y_0 \in A \wedge Y = [y_0]_S$$

$$\text{Since } (x, y) \in T \text{ then } (x_0, y_0) \in T$$

$$\text{which means } \exists x \in [x_0]_S \exists y \in [y_0]_S (xRy) \cdot \cdot$$

$$\text{Similarly, as } (Y, X) \in T \text{ so}$$

$$\exists y \in [y_0]_S \exists x \in [x_0]_S (yRx) \checkmark$$

$$\text{By reflexivity of } R, \ yRx \text{ as well so } xR^{-1}y$$

$$\text{as } S = R \cap R^{-1} \text{ so } xSy \text{ so } x \in [y]_S \text{ or } y \in [x]_S$$

$$\text{so } [x]_S = [y]_S \text{ which means } X = Y$$

We could also do this by applying our isomorphism separately. The key is to realize which elements belong to same class & so their classes are equal

Transitivity

$$\text{Supp } X \in A/S, Y \in A/S, Z \in A/S$$

$$\text{Supp } (X, Y) \in T \wedge (Y, Z) \in T$$

$$\longrightarrow (X, Z) \in T$$

$$\exists x \in A (X = [x]_S)$$

$$\exists x \in X \exists z \in Z (xRz)$$

$$\exists y \in A (Y = [y]_S)$$

$$\exists z \in A (Z = [z]_S)$$

$$(X, Y) \in T \text{ so } \exists x \in X = [x]_S \exists y \in Y = [y]_S (xRy)$$

$$\text{Similarly } \exists y \in Y = [y]_S \exists z \in Z = [z]_S (yRz)$$

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$\text{so let } x \in X \wedge y_1 \in Y (x R y_1)$
 $\text{and } y_2 \in Y \wedge z \in Z (y_2 R z)$
 $\text{since } y_1 \in y_2 \in [y]$
 $\text{so } y_1 S y \wedge y_2 S y$
 $\text{since } S = R \cap R^{-1}$
 $\text{so } y_1 R y \wedge y_2 R y$
 $\text{so } y R y_1 \wedge y R y_2$
 $\text{as } x R y_1 \in y_1 R y$
 $\text{so } x R y$
 $\text{as } x R y \wedge y R y_2 \text{ so } x R y_2$
 $\text{As } x R y_2 \wedge y_2 R z$
 $\text{so } x R z$

$\text{--- more easy check}$
 $\text{as } y_1 \in y_2 \in [y]$
 $\text{so } y_1 S y_2$
 $\text{so } y_1 R y_2$
 $\text{as } x R y_1 \text{ so}$
 $x R y_2$
 $\& \text{ as } y_2 R z$
 $\text{so } x R z$

Q26) $E = \{1, 2, \dots, 100\}$, $A = P(E)$

$R = \{ (x, y) \in A \times A \mid y \text{ has atleast as many elements as } x \}$
 $N(y) \geq N(x)$

(a) Reflexivity

clearly for any $(x, x) \in A \times A$, $N(x) \geq N(x)$ so $(x, x) \in R$.

Transitivity

supp $(x, y) \in R$ $\wedge$ $(y, z) \in R$

$(x, z) \in R$

so have $x \in A$

also $N(y) \geq N(x)$ & $N(z) \geq N(y)$

so clearly $N(z) \geq N(x)$ so $(x, z) \in R$

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(b) S is equivalence relation on A & $S = R \cap R^{-1}$

$$T = \{(x, y) \in A/S \times A/S \mid \exists x \in X \exists y \in Y (xRy)\}$$

$$R = \{(x, y) \mid N(y) \geq N(x)\}$$

$$R^{-1} = \{(y, x) \mid N(y) \geq N(x)\} = \{(x, y) \mid N(x) \geq N(y)\}$$

$S = R \cap R^{-1}$ so S contains only those pairs of set (x, y)

$$\text{such that } |x| = |y| \text{ so } S = \{(x, y) \in A/S \mid |x| = |y|\}$$

A/S contains those classes of which all elements (sets) have same no of elements. Along with an additional class $\{p\}$.

$$A/S = \{[x] \mid x \in A\} = \{\{y \mid |y| = |x|\} \mid x \in A\}$$

since there are 100 classes of elements from 1-100. And the class $\{p\}$, so A/S has 101 elements.

$$T = \{(x, y) \in A/S \times A/S \mid \exists x \in X \exists y \in Y (|x| \leq |y|)\}$$

To see that T is total order.

$$\text{Supp } x \in A/S \text{ \& } y \in A/S \longrightarrow xTy \vee yTx$$

if xTy then goal is clear.

$$\text{Supp } \neg xTy \text{ so } (x, y) \notin T \text{ so}$$

$$\forall x \in X \forall y \in Y (|x| > |y|)$$

but this means $|y| \leq |x|$

if this is true for all $x \in X$ then certainly it's true for atleast one $x \in X$ so yTx .

Note: The solution manual has described this in a better notation

by introducing $J = \{0, 1, \dots, 100\}$ & then $P_j = \{x \in A \mid x \text{ has } j \text{ elements}\}$

then $A/S = \{P_0, P_1, \dots, P_{100}\} = \{P_j \mid j \in J\}$

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$$U \subseteq A$$

$$\forall x \in U \forall y \in U (x \neq y \rightarrow x \cap y = \emptyset)$$

$$\forall x \in U (x \neq \emptyset)$$

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Q27) F is a partition of A .

F refines G if $\forall x \in F \exists y \in G (x \subseteq y)$

P = set of all partitions of A ($F, E, F', F_1, \dots$)

$R = \{(F, G) \in P \times P \mid F \text{ refines } G\}$

(a) Reflexivity:

supp $F \in P$

$\rightarrow (F, F) \in R$

$= F \text{ refines } F$

$= \forall x \in F \exists y \in F (x \subseteq y)$

supp $x \in F$ then let $y = x$. clearly $x \subseteq x = y$.

Antisymmetry.

supp $(F, G) \in R$ and $(G, F) \in R$ ~~$(G, F) \in R$~~ $F = G$

if F refines G iff $\forall x \in F \exists y \in G (x \subseteq y)$ — (1)

similarly $\forall x \in G \exists y \in F (x \subseteq y)$ — (2)

$\rightarrow x \in F$. Now by (1), we can conclude

$x \in G$

$\exists y \in G (x \subseteq y)$ — (3)

so from (2), we can conclude since $y \in G$

$\exists x_1 \in F (y \subseteq x_1)$ — (4)

from (3) & (4), we can also see that $x \subseteq x_1$

$x \in G (x \subseteq y)$, $x \in F$

$x_1 \in F (y \subseteq x_1)$, $x \subseteq x_1$

As $x \in F$ so $\exists x \in A (x = \{x\})$

As $x_1 \in F$ so $\exists x_1 \in A (x_1 = \{x_1\})$ so since $x \subseteq x_1$ so $\{x\} \subseteq \{x_1\}$

clearly $x \in \{x\}$ but as $\{x\} \subseteq \{x_1\}$ so $x \in \{x_1\}$

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so $\{x\} = \{x_1\}$ so $X = X_1$

Now as $Y \in G$ so $\exists y \in A (y = [y])$

as $Y \subseteq X_1$ so $[y] \subseteq [x_1]$

as $y \in [y]$ so $y \in [x_1]$ so $[y] = [x_1]$

so $Y = X_1$ but $X_1 = X$ so $Y = X$

& as $X \in F$ & $Y \in G$ but $Y = X$ so ~~$X \in G$~~ $X = Y \in G$

← Similar reasoning.

Transitivity

Supp $(F, G) \in R \wedge (G, H) \in R$

$(F, H) \in R$.

$\forall x \in F \exists y \in G (x \subseteq y)$ — (1)

$\forall x \in F \exists y \in H (x \subseteq y)$

$\forall x \in G \exists y \in H (x \subseteq y)$ — (2)

Supp $x \in F$

$\exists y \in H (x \subseteq y)$

by (1) ~~$\exists y_1 \in G$~~ $\exists y_1 \in G (x \subseteq y_1)$

by (2) $\exists y_2 \in H (y_1 \subseteq y_2)$

As $x \subseteq y_1$ & $y_1 \subseteq y_2$ so $x \subseteq y_2$

& clearly $y_2 \in H$ so let $y = y_2$ — our goal follows

(b) S & T are equivalence relations on A

$F = A/S, G = A/T$

$S \subseteq T \iff F$ refines G

→ Supp $S \subseteq T$

$\forall x \in F \exists y \in G (x \subseteq y)$

Supp $x \in F$

$\exists y \in G (x \subseteq y)$

then $\exists x \in A (x = [x])$

lets prove ← side first

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$$\forall x \in F \exists y \in G (x \leq y) \quad \text{---} \quad \text{---} \quad \text{---}$$

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← Suppose S satisfies G :

$$S \subseteq T$$

$$\text{supp}(a, b) \in S$$

$$(a, b) \in T$$

$$\text{then } a \in [b]_S$$

$$\text{so } [a]_S = [b]_S$$

$$\text{As } F = A/S \text{ so } [a]_S \in F \text{ \& } [b]_S \in F$$

$$\text{by } \textcircled{1} \exists Y_1 \in G ([a]_S \subseteq Y_1)$$

$$\exists Y_2 \in G ([b]_S \subseteq Y_2)$$

$$\text{which means as } Y_1, Y_2 \in G = A/T$$

$$\exists [y_1] \in G ([a]_S \subseteq [y_1]_T)$$

$$\exists [y_2] \in G ([b]_S \subseteq [y_2]_T)$$

$$\text{as } [a]_S = [b]_S \text{ so } [a]_S \subseteq [y_2]_T$$

$$\text{as } a \in [a]_S \text{ so } a \in [y_2]_T \therefore \text{subset}$$

$$\text{so } [a]_S = [y_2]_T$$

$$\text{It is clear by now that } [a]_S = [b]_S = [y_1]_T = [y_2]_T$$

$$\text{Now clearly } (a, y_1) \in T \text{ as } a \in [y_1]_T$$

$$\& \text{ } (y_1, b) \in T \text{ as } (y_1, b) \in T \therefore (a, b)$$

$$\& (y_1, b) \in T \text{ so by transitivity of } T$$

$$(a, b) \in T$$

← Suppose $S \subseteq T$

$$\forall x \in F \exists y \in G (x \leq y)$$

$$\text{supp } x \in F \text{ then } \exists x \in A (x = [x]_S)$$

$$\exists y \in G (x \leq y)$$

$$F = A/S \therefore G = A/T$$

$$\text{As } x \in [x]_S \text{ so } (x, x) \in S \text{ as } S \subseteq T \text{ so } (x, x) \in T$$

$$\text{so } x \in [x]_T \text{ so } [x]_S \subseteq [x]_T (\because \text{we can separately prove this})$$

$$\text{clearly } [x]_T \in G \text{ as } [x]_S \subseteq [x]_T \text{ as } [x]_S \subseteq [x]_T$$

$$\text{so let } y = [x]_T \& \text{ goal follows}$$

$$\forall y \in A (xRy) \wedge \forall x' [\forall y \in A (x'Ry) \rightarrow x'Rx]$$

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(c) F & G are partitions of A To prove

$F \cdot G$ is glb of set $\{F, G\}$ in R

First we try to prove that $F \cdot G$ is lower bound of $\{F, G\}$

or $\forall H \in \{F, G\} ((F \cdot G) R H)$ or $(F \cdot G, H) \in R$

supp $H \in \{F, G\}$ $(F \cdot G, H) \in R$

$$\forall x \in F \cdot G \exists y \in H (x \leq y)$$

$$\text{supp } x \in F \cdot G \quad \exists y \in H (x \leq y)$$

iff $x \in P(A) \wedge x \neq \emptyset$

$$\wedge \exists y \in F \exists z \in G (x = y \cap z)$$

so ~~$\forall y \in F \wedge \forall z \in G (x = y \cap z)$~~

$$y_0 \in F \quad z_0 \in G \quad (x = y_0 \cap z_0)$$

Note: I found y by considering an example of simple case. I noted

that $y_0 \cup z_0, y_0 \cap z_0, y_0 \setminus z_0, y_0^c \cap z_0$ cannot be required set by

seeing contradictions. The right set is y_0 for case when $H = F$.

as $x = y_0 \cap z_0$ so $x \leq y_0 \cap z_0$ & $y_0 \cap z_0 \leq x$. as $x \leq y_0 \cap z_0$ so

$x \leq y_0 \wedge x \leq z_0$. & clearly $y_0 \in F$. similar reasoning for $H = G$

(case 1) $H = F$

let $y = y_0$ then as $y_0 \in F$ & $x = y_0 \cap z_0$ so $x \leq y_0$. Good

follows

(case 2) $H = G$ has similar reasoning.

Now we prove that $F \cdot G$ is g.l.b of $\{F, G\}$

$$\forall J \in P [\forall H \in \{F, G\} (J, H) \in R \rightarrow (J, F \cdot G) \in R]$$

let $J \in P$ be arbitrary

supp $\forall H \in \{F, G\} (J, H) \in R$

some $F \in \{F, G\}$ so $(J, F) \in R$

$$\left(\begin{array}{l} \forall x \in J \exists y \in F \cdot G (x \leq y) \\ (J, F \cdot G) \in R \end{array} \right)$$

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$$\text{iff } \forall X \in \mathcal{S} \exists Y \in \mathcal{F} (X \subseteq Y) \quad \text{--- (1)}$$

Similarly $(\mathcal{F}, \mathcal{G}) \in R$

$$\text{so } \forall X \in \mathcal{S} \exists Y \in \mathcal{G} (X \subseteq Y) \quad \text{--- (2)}$$

This can easily be seen as $\mathcal{S}$ is l.b for $\{\mathcal{F}, \mathcal{G}\}$
so $\mathcal{S}$ refines both $\mathcal{F}$ & $\mathcal{G}$

Let $X \in \mathcal{S}$ be arbitrary

Since $X \in \mathcal{S}$

$$\text{by (1)} \quad Y_2 \in \mathcal{F} (X \subseteq Y_2)$$

$$\text{by (2)} \quad Y_3 \in \mathcal{G} (X \subseteq Y_3)$$

$$\text{so } X \subseteq Y_2 \cap Y_3.$$

$$\exists Y \in \mathcal{F} \cap \mathcal{G} (X \subseteq Y)$$

$$\left[\exists Y \left[Y \in \mathcal{P}(A) \wedge Y \neq \emptyset \wedge \right. \right.$$

$$\left. \exists Y_1 \in \mathcal{F} \exists Y_2 \in \mathcal{G} (Y = Y_1 \cap Y_2) \right]$$

$$(X \subseteq Y)$$

so let $Y = Y_2 \cap Y_3$. Goal follows. As $Y_2 \in \mathcal{F}$ so $Y_2 \subseteq A$
& $Y_3 \subseteq A$. so $Y_2 \cap Y_3 \subseteq A$ so $Y_2 \cap Y_3 \in \mathcal{P}(A)$

& since $Y \in \mathcal{F} \cap \mathcal{G}$. Y belongs to a partition so $Y = Y_2 \cap Y_3 \neq \emptyset$.
clearly $Y_2 \in \mathcal{F}$ & $Y_3 \in \mathcal{G}$. $\exists X \subseteq Y_2 \cap Y_3$.

so $\mathcal{F} \cap \mathcal{G}$ is the g.l.b of $\{\mathcal{F}, \mathcal{G}\}$ with respect to partial order R .